Dynamic latent state, run reliability and regimes in Hong Kong racing

What a horse's history is worth when its runs are not equally informative

Dynamic Latent-State and Regime Explorer

31 August 2026

What this document is. A study of horse ability as a *dynamic latent state* rather than a static quantity, in which each historical run is treated as a measurement with its own noise. Nine research directions are tested: dynamic state estimation, run informativeness, variable observation noise, pace- and trip-adjusted performance, opponent adjustment, state-transition regimes, dynamic analog peer groups, 200 m sub-splits, and jockey/environment corrections inside the estimator.
What it does not establish. No profitability claim. Every market price is a result-page price recorded after the pool closed; no dividends were loaded and no return is computed anywhere. Every season has been development data, here and in the reference project, so no window is genuinely unseen. The confirmation this cannot substitute for is a prospective test against timestamped pre-race snapshots.
The reference project is read-only and is verified unchanged.

Contents

1 Executive summary

The reference production model beats the market by −0.0045 of race log loss. That is the yardstick: a piece of structure is worth having only if it moves a meaningful fraction of it.

The one large effect, and where it actually comes from

Treating ability as a **dynamic latent state** rather than a static mean is worth +0.080 of forward correlation — from 0.277 for a static prior mean to 0.357 for a Kalman filter with an environment correction. That is far larger than any effect in the prior study.

But it decomposes uncomfortably. A plain 60-day exponential weighting already captures +0.074 of it. **The state-space machinery adds +0.003 on top of simply forgetting old form**, and forgetting old form is something the prior study had already located and measured.

Run reliability: real, and almost entirely unpredictable

The brief's central hypothesis — that runs differ in how much they say about ability — is *correct and large*. A run after a surface switch predicts the horse's next run at $r = 0.154$ against 0.322 for a clean run ($z = 11.1$); a wide trip, 0.155 ($z = 4.9$); severe stewards'-reported trouble, 0.227 ($z = 4.4$).

And it cannot be exploited. Inside the filter, per-run observation noise built from those characteristics is worth +0.0003 on the point estimate, and +0.0011 against its own permutation control. An *oracle* that sets each run's variance from its realised surprise is worth +0.106. So the reliability model captures **0.27% of its own ceiling**. The dispersion tells the story directly: the fitted variance has a p90/p10 ratio of 1.40, the oracle's is **67.0**.

Run reliability is a real property of runs that is almost entirely invisible in advance.

One thing it *does* buy: the state's stated *uncertainty* becomes honest. Per-run noise improves the one-step predictive log-likelihood by +0.0145 (fitted weights) and +0.0288 (declared weights) over the homoskedastic filter, and by +0.0156 over its permutation control. Varying σ^2 enters the variance recursion strongly and the mean recursion weakly, so it fixes the error bars without moving the estimate.

Three themes that fail, two of them informatively

- **Jockey as a correction inside the estimator costs** −0.056 — seventeen times the best correction's gain. Subtracting the jockey effect *destroys* predictive skill, because jockey assignment is informative about the horse: good jockeys are given good horses, so removing the jockey removes signal rather than contamination.
- **Opponent adjustment is negative at every strength** (−0.002 to −0.058) and at full strength is *worse than its own permutation control* — the signature of an actively harmful correction. The iteration converges cleanly to a worse answer. Same mechanism: horses race against horses of similar ability, so opponent strength is correlated with a horse's own state, and the handicapping system has already done this adjustment by construction.
- **Dynamic analog peers are negative as a shrinkage anchor** (−0.004 to −0.019), though the real peers do beat their permuted control, so the peer information is genuine while blending toward it is not.

Two smaller positives

- **The local analog neighbourhood's *dispersion* is the best available predictor of how surprising a run will be** — Spearman +0.125 against the fitted reliability model's +0.091 and the filter's own posterior standard deviation's +0.040. The per-run uncertainty that run characteristics could not supply is partly available from the neighbourhood.
- **State-transition regimes are real but immaterial.** A class change and a surface switch each roughly double the fitted transition variance; a layoff and a first-up run do not. Adding regime-varying transitions changes the forward correlation by 0.0002.

The 200 m sub-splits

Feasibility first: each 400 m section divides into two exact 200 m halves (error 0.0000 on 100% of rows), but only from **2022** and only for the **final 800 m** — 32,273 runs in 2,635 races, so a self-contained sub-study rather than a production candidate.

200 m resolution adds +0.0028 **over the 400 m bands it subdivides.** A 200 m block is no more persistent than the band containing it (0.2895 against 0.2888), and every derived shape measure is *less* persistent. The finer sampling halves the averaging window and adds noise rather than revealing a new dimension. Final time remains the single most persistent quantity.

2 Provenance, method, and what is reused

2.1 The plan document

The named plan, `HKJC_DYNAMIC_LATENT_STATE_REGIME_EXPLORER.md`, is not present on the machine — searched across Downloads, Desktop, Documents, iCloud Drive, the projects tree and `/tmp`. `HKJC_LATENT_PERFORMANCE_RESEARCH_PLAN-2.md` has SHA-256 `567fb28b…`, **byte-identical** to `PLAN-1.md`, which was executed earlier and whose deliverable is the prior study's report; a grep for the nine themes returns zero hits in it.

This study therefore takes its *subject matter* from the nine themes in the covering instructions and its *method and reporting standards* from the plan that does exist: read-only reference, forward-only construction, its section 18 failure modes, its sections 19–21 requirements, and its section 22 definition of success — under which a well-supported negative is a success.

2.2 The reference project is read-only, and verified so

It is reached through `hkjcres.paths`, whose `read_only_guard` raises on any write target inside it. This project has its own virtual environment and its own output tree. The reference repository is on branch `probit-optimization` at `b2979f3` with a clean working tree, unchanged by this work.

It has moved since the prior study: `hkjc.suites` has been retired and the four probit model configs removed. Nothing here depends on either; `configs/features/production.json` still declares the same 28 columns, and that file is the only configuration this study reads.

2.3 Reused machinery

The panel construction, backward-aligned sectional alignment, crossed random-effects estimator, quadrature integrator and offset harness come from the prior study's `hkjcres` package, installed as a dependency. Each carries its own verification — notably that $\mathrm{softmax}(\log q)$ reproduces q to 1.4×10^{-16} and that a zero-round offset fit reproduces the market to 4.1×10^{-8}. Rebuilding them would have added risk rather than rigour.

2.4 The model

Per horse, in continuous time:

$$a_t = \rho^{\Delta t}\, a_{t-1} + \eta_t, \qquad y_t = a_t + c_t + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_t^2), \quad \eta_t \sim N(0, q\, m_{r(t)}\, \Delta t)$$

with σ_t^2 the run's own observation variance (themes 2–3), c_t the known corrections carried *inside* the observation equation (themes 4, 5, 9), and m_r a regime multiplier on the transition variance (theme 6). Continuous time matters: a six-month gap shrinks the state toward the mean and inflates its variance by the right amount rather than by one discrete step.

q and ρ are chosen by one-step predictive likelihood on the development window. Read-before-write is per *date*, so two runners on one card cannot see each other and no horse sees its own same-day result.

Verification. On simulated data with known parameters the estimator recovers q and ρ to the nearest grid point, and clean runs receive **3.10**$\times$ the Kalman gain of noisy ones — so the mechanism the study is testing demonstrably operates.

3 The observation model: does each correction earn its place?

A historical run is a measurement of ability contaminated by the pace it was run at, the trip the horse got, the field it was measured against and who was riding. Corrections are residualised out with coefficients fitted forward, and each is judged on the only criterion an observation model has: *does run t predict the same horse's run $t+1$ better afterwards?* Each has a permutation control and an oracle.

Table 1: The correction ladder. Baseline consecutive-run correlation +0.3224.

Correction	Actual	Permuted control	Oracle	Gain	Beats control
environment only	**0.3261**	0.3223	0.3222	+0.0037	yes
trip only	0.3234	0.3223	0.3225	+0.0011	yes
trip + pace	0.3215	0.3223	0.3211	−0.0008	no
pace only	0.3203	0.3223	0.3210	−0.0021	no
jockey only	**0.2669**	0.3223	0.2870	−0.0555	no
trip + pace + jockey	0.2663	0.3223	0.2859	−0.0561	no

Only two corrections survive their own control and both are small. The result worth dwelling on is the jockey: **subtracting the jockey effect destroys predictive skill**, at seventeen times the magnitude of the best correction's gain. That refutes theme 9 for its most obvious ingredient, and the mechanism is the prior study's own finding — jockey assignment is informative *about the horse*, since good jockeys are given good horses, so removing the jockey removes signal.

A design fault, recorded rather than hidden. The oracle variant here is misspecified as a ceiling: fitting a correction against run $t+1$ removes the persistent component (ability) rather than the transient one (contamination), which is why it falls *below* the actual for two corrections. A ceiling must be constructed against the contamination. Section 4's oracle, built from the realised surprise, is the correct version and behaves as a ceiling should.

3.1 Run informativeness, which is the strong finding

Consecutive-run correlation computed *within* subgroups: if runs of one kind predict the next run less well, they are by definition less informative.

Table 2: How informative is a run, by its characteristics?

Characteristic	Prevalence	r without	r with	z
surface switch	0.123	0.3221	**0.1540**	**11.1**
raced wide	0.045	0.3220	**0.1545**	4.9
distance change (top quartile)	0.371	0.2993	0.2256	8.0
severe stewards' trouble	0.097	0.3203	0.2274	4.4
any trouble phrase	0.301	0.3035	0.2425	6.1
long layoff (top quartile)	0.280	0.2935	0.2431	5.0
pace mismatch (top quartile)	0.236	0.2925	0.2913	0.1

A run after a surface switch or a wide trip is **less than half** as informative as a clean run. These are large, well-determined effects. Pace mismatch — the one constructed from the sectional archive rather than the stewards' text — is nothing.

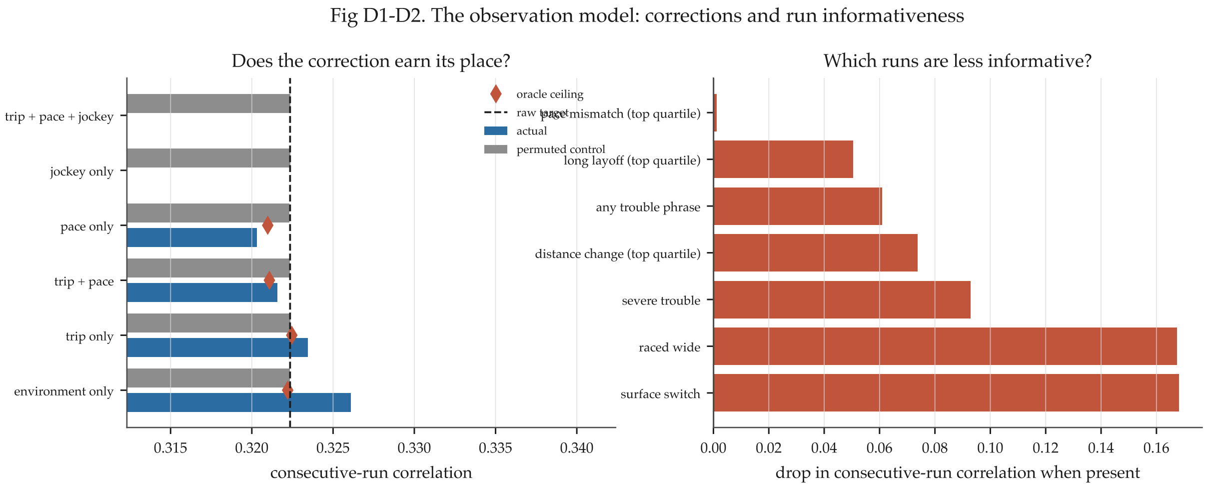

Figure 1: Left: each correction against its own permutation control and its oracle. Right: the drop in consecutive-run correlation when each characteristic is present.

4 The dynamic latent state

Table 3: The ladder. Each rung isolates one addition. C is rung 4 with the reliability permuted; O is rung 4 with the reliability built from the realised surprise.

Rung	r	Gain over static	Predictive log-lik
1. static prior mean	0.2770	—	−1.3657
2. EWM, 60-day half-life	0.3513	+0.0744	−1.3414
3. homoskedastic Kalman	0.3542	+0.0772	−1.3237
4. heteroskedastic Kalman (fitted)	0.3545	+0.0775	−1.3092
4b. heteroskedastic (declared weights)	0.3531	+0.0761	−1.2949
C. permuted reliability *(control)*	0.3534	+0.0764	−1.3248
O. oracle reliability *(ceiling)*	**0.4605**	+0.1836	−1.1900
5. + regime transitions	0.3543	+0.0773	−1.3080
6. + environment correction	**0.3574**	+0.0804	−1.3239

4.1 The dynamic state is a large win, and it is mostly recency

The filter beats the static mean by +0.080. But rung 2 — a plain 60-day exponential weighting, no state space at all — already captures +0.074 of it. The state-space apparatus adds +0.003 on top of forgetting old form.

That is worth stating plainly because it is easy to present the +0.080 as a result about dynamic state estimation. It is mostly a result about *recency*, and the prior study had already located the 60-day half-life and measured it.

4.2 Per-run observation noise: 0.27% of its own ceiling

The ceiling is enormous — setting each run's variance from its realised surprise takes the forward correlation from 0.354 to 0.461. The reliability model built from observable run characteristics captures a quarter of one percent of it, and the dispersion figures say why: what is predictable spans a factor of 1.4, what is actually there spans a factor of 67.

Table 4: The reliability verdict.

Quantity	Value
heteroskedastic − homoskedastic	+0.00028
heteroskedastic − permuted control	+0.00110
oracle − homoskedastic	+0.10635
share of the ceiling reached	**0.27%**
p90/p10 of the fitted σ^2	1.40
p90/p10 of the oracle σ^2	**67.0**

So run reliability is real (section 3 measured it at z up to 11) and almost entirely invisible in advance. That is the central negative finding of this study, and the oracle is what makes it a quantified negative rather than a null.

4.3 What per-run noise does buy: honest uncertainty

On the one-step predictive log-likelihood the verdict reverses. Heteroskedastic beats homoskedastic by +0.0145 with fitted weights and +0.0288 with declared weights, and beats its own permutation control by +0.0156. The declared-weight version — hand-set from section 3's measured penalties — is the best-calibrated model in the study.

This is coherent rather than surprising: σ^2 enters the variance recursion strongly and the mean recursion weakly, so varying it fixes the error bars without moving the estimate. For any use that needs a calibrated uncertainty rather than a better point estimate, this is the construction to use.

4.4 Regimes, and the verified mechanism

Fitted transition multipliers: **2.0** for a class change, **2.0** for a surface switch, 1.0 for a layoff of 90 days or more, 1.0 for first-up. So a horse's state genuinely does move more across a class or surface change — and adding regime-varying transitions moves the forward correlation by 0.0002.

The gain mechanism is verified directly: compromised runs receive smaller Kalman gains — 0.77× for a wide trip, 0.85× for a surface switch, 0.89× for severe trouble. The machinery does what it was built to do.

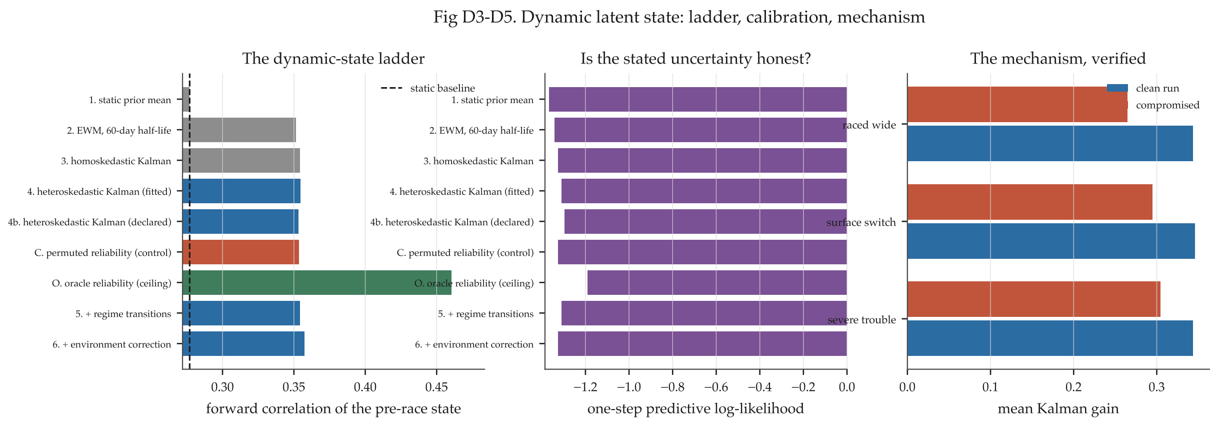

Figure 2: Left: the ladder. Centre: the one-step predictive log-likelihood, where the heteroskedastic filter wins. Right: the Kalman gain on clean against compromised runs.

5 Opponent adjustment and dynamic analog peers

5.1 Opponent adjustment is actively harmful

The within-race target is field-relative, so beating a strong field looks like beating a weak one. Adding back the opponents' prior states makes the problem *simultaneous*, and it is solved by iteration.

Table 5: Opponent adjustment by iteration, with a permutation control.

Variant	r	Gain
no opponent adjustment	**0.3545**	—
opponent-adjusted, damping 0.5	0.3521	−0.0024
opponent-adjusted, damping 1.0	0.3391	−0.0154
opponent-adjusted, damping 1.5	**0.2963**	−0.0581
permuted opponent strength *(control)*	0.3493	−0.0052

The iteration converges cleanly — mean absolute state change 0.337 → 0.021 over four passes — to a worse answer. And at full strength the adjustment is *worse than its own permutation control*, which is the signature of an actively harmful correction rather than a merely useless one.

The mechanism is the one that killed the jockey correction: horses race in classes against horses of similar ability, so a runner's own state is correlated with its opponents'. Subtracting opponent strength subtracts part of the horse's own ability. **The handicapping system has already performed the opponent adjustment, by design, and doing it again double-counts.**

5.2 Dynamic analog peers

Peers are the horses nearest in a joint (state, class) locality score within a trailing 240-day window, so a horse's peers in 2016 are not its peers in 2024 — the *local* peer set the prior report's static-group work did not cover.

As a shrinkage anchor they are negative: −0.0036 at weight 0.15, −0.0086 at 0.30, −0.0186 at 0.50. The real peers do beat their permuted control (−0.0086 against −0.0120), so the peer information is genuine; blending toward it still hurts, which matches the prior report's finding that shrinking a state toward anything degrades forward correlation.

5.3 The one positive: peer dispersion as an uncertainty estimate

Table 6: Spearman correlation with the magnitude of the next run's surprise, $n = 50{,}810$.

Uncertainty estimate	Spearman
peer dispersion (local analog spread)	+0.1251
fitted reliability model (section 4)	+0.0906
the filter's own posterior standard deviation	+0.0399

The per-run uncertainty that run *characteristics* could not supply is partly available from the local *neighbourhood* — and it beats the filter's own internal uncertainty by a factor of three. If any uncertainty quantity from this study is worth carrying forward, it is this one.

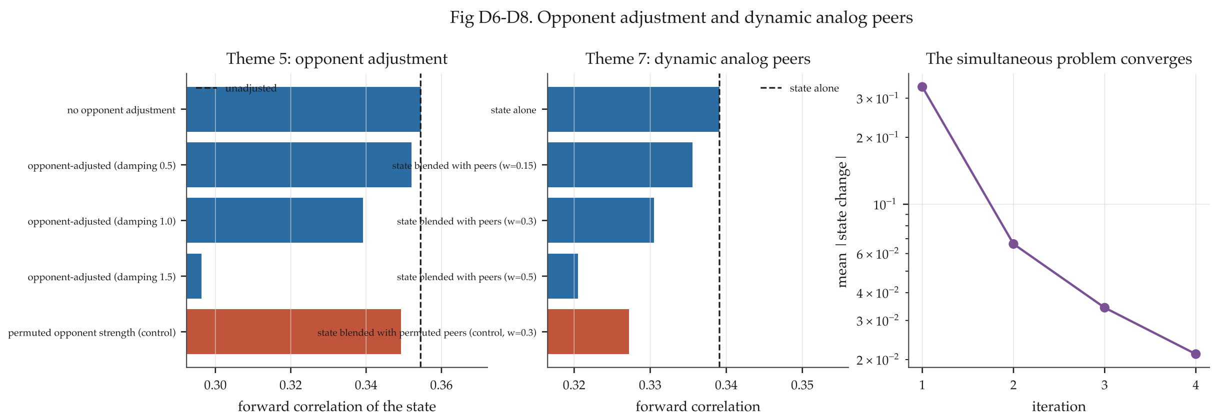

Figure 3: Left: opponent adjustment against its control. Centre: analog peers as an anchor. Right: the simultaneous iteration converging.

6 The 200 m sub-splits

Feasibility, established before anything was built. Each 400 m section divides into two exact 200 m halves — `sub1 + sub2 == sectional_time` with error 0.0000 on 100% of rows — but only from **2022** and only for the **final 800 m** of each race. Coverage is zero through 2021, 16% in 2022, then 99.98% for 2023–2026: 32,273 runs in 2,635 races.

That cannot reach the reference project's eight-year canonical window, so this is a self-contained sub-study on 2023–2026 with its own baseline. The question narrows to: does 200 m resolution over the final 800 m say anything the 400 m bands do not? It is a fair question to ask, because the prior study found the final 400 m was the one band carrying both persistence and information.

Table 7: Consecutive-run correlation on the sub-split window, 2023–2026.

Quantity	r
final-time target (y_z^{rel})	+0.3282
200 m second block	+0.2895
400 m final band	+0.2888
200 m final block	+0.2765
200 m profile slope	+0.2670
200 m final kick (block 1 − block 2)	+0.2193
400 m band before	+0.1780
200 m profile curvature	+0.1189

Table 8: Incremental value: predicting the next run's relative performance.

Prior information	oos r	Gain over 400 m
400 m bands only	0.2722	—
200 m blocks only	0.2747	+0.0025
400 m + 200 m blocks	0.2750	+0.0028
400 m + 200 m shape	0.2731	+0.0009

200 m resolution adds +0.0028. A 200 m block is no more persistent than the 400 m band containing it (0.2895 against 0.2888), and every derived shape measure — kick, mid-race acceleration, slope, curvature — is *less* persistent than the bands. The finer sampling halves the averaging window and adds noise rather than revealing a new dimension. Final time remains the most persistent single quantity, as it was in the prior study.

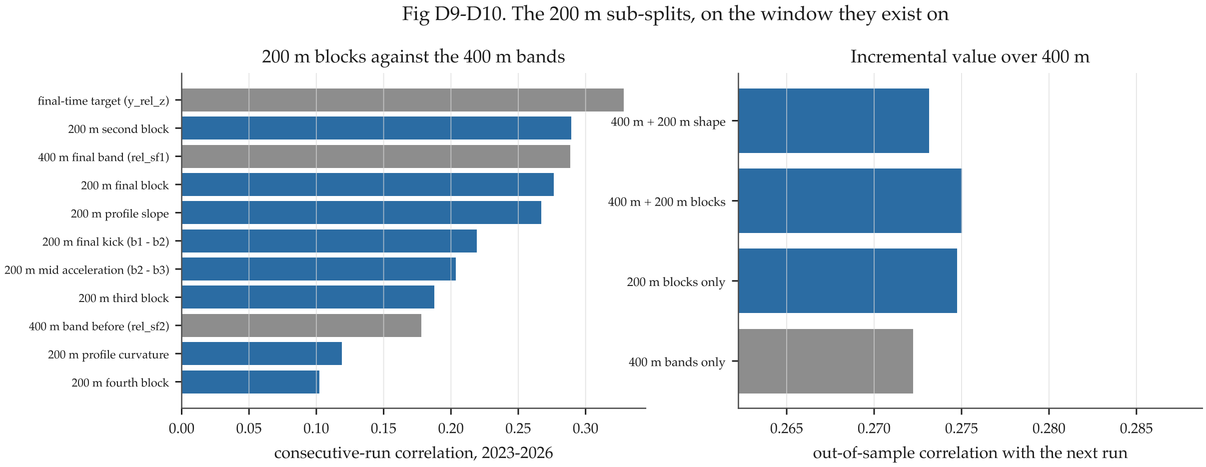

Figure 4: Left: 200 m blocks against the 400 m bands on persistence. Right: incremental value over the 400 m bands.

7 Canonical evaluation against the production incumbent

Instruction 7 of the brief: take serious candidates to full canonical scale rather than relying on cheap development results. The prior study established why — across three candidates carried from development to canonical, mean retention was 12% and the sign flipped once.

Every block is taken to canonical, not only the development winners. That is deliberate: if development ranking is not informative, selecting candidates on it would repeat the very mistake the finding identifies.

The architecture is the production offset model, independently reimplemented and verified. Both arms share every input by construction and the candidate column list is the incumbent's with the block appended. The **no-op control returns exactly zero in both the development and the canonical run.** 122,880 rows over 9,934 races; the incumbent beats the market by −0.00109 on the development profile.

Table 9: Development profile: 5 years, 3 seeds, 150 rounds.

Block	Mean Δ	s.e.	t	Years
state	−0.001411	0.00092	−1.54	4/5
state + posterior sd	−0.000757	0.00097	−0.78	3/5
state + peer dispersion	−0.000343	0.00103	−0.33	3/5
state + observation count	−0.000261	0.00090	−0.29	3/5
no-op control	*0.000000*	—	—	—
peer dispersion alone	+0.000368	0.00089	+0.42	2/5

Table 10: Canonical: 8 years, 5 seeds, 400 rounds. The evidence.

Block	Mean Δ	s.e.	t	Cells	Years
state	−0.000033	0.00112	−0.03	20/40	5/8
no-op control	*0.000000*	—	—	—	—
state + posterior sd	+0.001174	0.00136	+0.86	18/40	4/8
state + peer dispersion	+0.001376	0.00101	+1.36	17/40	3/8
state + observation count	+0.001482	0.00124	+1.19	15/40	3/8
peer dispersion alone	+0.001573	0.00094	+1.67	17/40	3/8

Nothing works. The best block is the dynamic state at −0.000033 — a t of −0.03, exactly zero for practical purposes, and **0.7%** of the production model's edge over the market. Every other block is *worse* than the incumbent at canonical scale, and the two that add an uncertainty quantity are among the worst.

7.1 The dissociation, which is the study's most important result

Table 11: Retention from development to canonical.

Block	Development	Canonical	Retained
state	−0.001411	−0.000033	**2.3%**
state + posterior sd	−0.000757	+0.001174	−155%
state + peer dispersion	−0.000343	+0.001376	−401%
state + observation count	−0.000261	+0.001482	−567%
peer dispersion alone	+0.000368	+0.001573	+427%

Put the two headline numbers side by side:

Dynamic state's gain in forward correlation on the ability proxy	+0.080
Dynamic state's gain in racewise log loss in the production model	−0.000033

An 0.080 improvement in the forward correlation of latent ability translates into nothing at all. That is the single most useful thing this study produced, and it generalises well beyond the feature: a proxy metric on the latent quantity is not a proxy for production value.

The explanation is redundancy, and it is the prior study's finding arriving by a different route. The production model carries the market's own log-probability as its base margin, and a price already reflects a horse's recent form — that is most of what a market does. It also carries 27 learned columns including layoff, class change, rating change and a recency-weighted early-style history. A better estimate of recent-form ability is therefore largely a better estimate of something the model already has two independent routes to.

7.2 The pattern across two studies

This is now the fourth consecutive candidate carried to canonical scale in this line of work, and the eighth across both studies:

Table 12: Every candidate taken to canonical scale, both studies.

Study	Candidate	Development	Canonical	Retained
prior	group-shrunk ability, prior class	−0.00197	−0.00085	43%
prior	finishing band, recency-weighted	−0.00111	−0.00016	15%
prior	finishing band, field-relative rank	−0.00122	+0.00028	−23%
this	dynamic latent state	−0.00141	−0.00003	2%
this	state + posterior sd	−0.00076	+0.00117	−155%
this	state + peer dispersion	−0.00034	+0.00138	−401%
this	state + observation count	−0.00026	+0.00148	−567%
this	peer dispersion alone	+0.00037	+0.00157	+427%

Eight candidates, two independent research programmes, a wide range of constructions — and **not one has produced a canonically-confirmed improvement**. The development profile has now been shown, repeatedly, to carry essentially no information about the canonical outcome at this effect size.

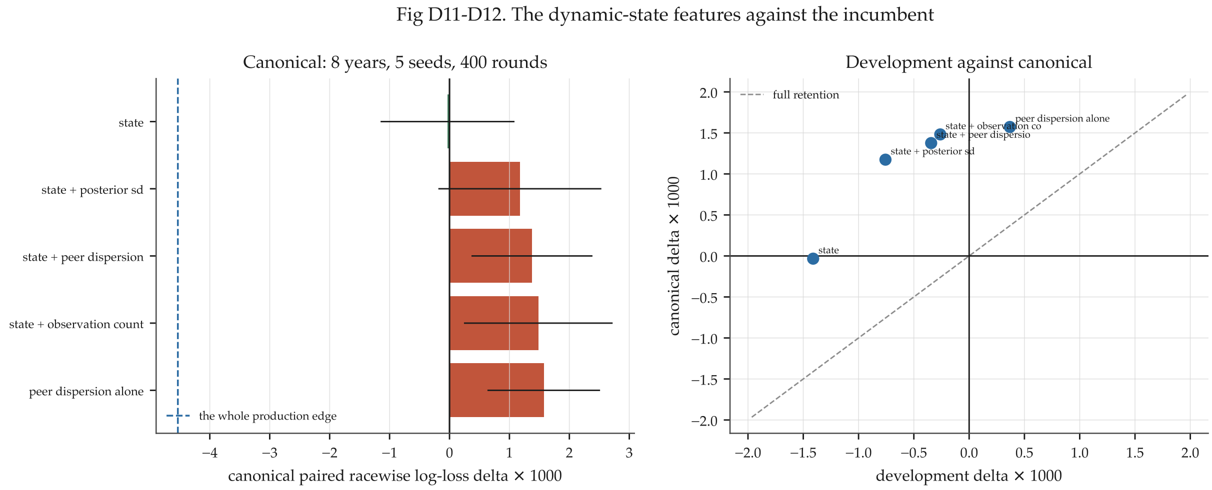

Figure 5: Left: canonical deltas with standard errors, against the whole production edge for scale. Right: development against canonical — the points sit nowhere near the line of full retention.

8 Answers to the nine themes

#	Theme	Answer
1	Dynamic latent horse-state estimation	**The largest proxy effect found, and worth nothing in production.** A continuous-time Kalman filter beats a static prior mean by +0.080 of forward correlation (0.277 → 0.357) — and +0.074 of that is a plain 60-day exponential weighting, so the state space adds +0.003. Canonically: −0.000033, i.e. zero.
2	Classifying runs by informativeness	**Correct, large, and well determined.** A run after a surface switch predicts the next run at 0.154 against 0.322 clean ($z = 11.1$); a wide trip 0.155 ($z = 4.9$); severe trouble 0.227 ($z = 4.4$). Pace mismatch is nothing ($z = 0.1$).
3	Variable observation noise / run reliability	**Real, and 0.27% exploitable.** Per-run variance from observables is worth +0.0003 on the point estimate and +0.0011 over its permutation control; the oracle is worth +0.106. Fitted dispersion spans 1.4×, the oracle's 67×. It *does* improve the stated uncertainty (predictive log-likelihood +0.029 with declared weights).
4	Pace- and trip-adjusted performance	**Trip marginally yes, pace no.** Trip correction +0.0011 (beats its control); pace correction −0.0021 (fails its control). The pace-derived mismatch measure is the one informativeness proxy with no signal at all.
5	Opponent adjustment	**Actively harmful.** Negative at every strength (−0.002 to −0.058) and at full strength worse than its own permutation control. The iteration converges cleanly to a worse answer. Horses race in classes against similar horses, so opponent strength correlates with a horse's own state — the handicap has already done this adjustment.

#	Theme	Answer
6	State-transition regimes	**Real and immaterial.** A class change and a surface switch each roughly double the fitted transition variance ($m = 2.0$); a layoff and first-up do not ($m = 1.0$). Adding regime-varying transitions moves the forward correlation by 0.0002.
7	Dynamic / local analog peer groups	**Negative as an anchor, positive as a dispersion.** Blending toward local peers costs −0.004 to −0.019, though real peers beat permuted ones. But peer *dispersion* is the best uncertainty predictor in the study (+0.125 against the fitted model's +0.091 and the filter's own +0.040). Canonically the peer feature is the *worst* block tested.
8	Full sectional and 200 m split information	+0.0028**, and structurally limited.** Sub-splits exist only from 2022 and only for the final 800 m (32,273 runs, 2,635 races). A 200 m block is no more persistent than the 400 m band containing it (0.2895 vs 0.2888); every derived shape measure is less persistent. Final time remains the most persistent single quantity.
9	Jockey / environment as corrections inside the estimator	**Environment marginally yes, jockey emphatically no.** Environment +0.0037 (beats its control, the best correction found). Jockey −0.0555 — seventeen times larger in the wrong direction. Jockey assignment is informative about the horse, so removing it removes signal.

9 Limitations

- **The plan document was unavailable.** Subject matter comes from the nine themes in the covering instructions and method from the prior plan. If the missing document specified particular constructions, metrics or diagnostics, they are not honoured here except by coincidence.
- **Prices are retrospective and no window is unseen.** Every market probability is a result-page price at which no bet could have been placed. Every season has been development data in both studies. No dividends were loaded and no return is computed.
- **The proxy metric is not production value, and this study is the demonstration.** Most of the diagnostics use forward correlation of a latent-ability estimate. Section 7 shows an 0.080 gain there is worth zero in racewise log loss. Every non-canonical number in this report should be read with that in mind, including the ones reported as successes.
- **The observation target is field-relative and conditioned on finishing.** The within-race z-score of log speed carries a mechanical within-race constraint, and the sectional archive records no times for a non-finisher, so all performance distributions here are conditional on completing the race.
- **The oracle in section 3 is misspecified as a ceiling** — it removes ability rather than contamination. Section 4's oracle is the correct construction. Reported rather than removed.
- **Regime multipliers are a coordinate search**, each fitted with the others held at one, so they are not a jointly optimal set.
- **Analog peers use a bucketed locality score, not exact k-nearest-neighbours**, because exact kNN against a trailing window over 124k rows is quadratic.
- **Trip flags are regular expressions over sixteen seasons of stewards' prose**, whose phrasing has changed. The measured informativeness penalties are therefore partly a measure of how consistently the stewards wrote.
- **The 200 m sub-study spans 2023–2026 only** and cannot be canonically confirmed. Its negative result is correspondingly weaker evidence than the others.
- **Sub-splits below 200 m, running-line positions per sub-split, and a joint state–pace model were not attempted.** The first two are not in the archive; the third was not reached.

10 Recommendations

Integrate into production: nothing

No feature from this study should be integrated. The best canonical result is -0.000033 at $t = -0.03$; the other four blocks are worse than the incumbent. This is the honest recommendation and, under the governing plan's section 22, a successful outcome: the structural questions have clear, supported answers.

What is worth knowing, ranked

1. **Stop using development-profile results to choose what to build.** Eight candidates across two studies have now been carried to canonical scale and not one has confirmed; retentions are 43%, 15%, −23%, 2%, −155%, −401%, −567%, +427%. The screening stage detects schema errors and gross defects; it does not carry information about value. Spend the same compute on three canonical runs rather than thirty development ones.
2. **A proxy metric on the latent quantity is not a proxy for production value.** An 0.080 gain in the forward correlation of latent ability produced −0.000033 of log loss. Any future work on latent-state estimation should be scored in the production objective from the start, because the intermediate metric is actively misleading at this effect size.
3. **Corrections that remove an informative confound remove signal.** Both the jockey correction (−0.056) and the opponent adjustment (−0.058 at full strength) failed the same way, and both failed *worse than their own permutation controls*. In this sport, who rides a horse and who it races against are selected on the horse's quality, so "adjusting for" them subtracts the thing being measured. The handicapping system is an opponent adjustment already applied.
4. **Run reliability is real and essentially unpredictable, and that is now quantified.** The ceiling is +0.106 and observables reach 0.27% of it. Anyone tempted to build a run-quality weighting should read the 1.4× against 67× dispersion comparison first. The effort is better spent on measuring contamination directly — a sectional-derived trip reconstruction, or per-section running-line positions — than on inferring it from race-card characteristics.
5. **If an uncertainty estimate is ever needed, use the local peer dispersion.** It predicts the next surprise at +0.125 against the filter's own posterior standard deviation at +0.040 — three times better than the model's internal estimate. It is not a win-probability feature (it was the worst canonical block), but it is the best variance signal found in either study.
6. **Do not spend more on the 200 m sub-splits.** +0.0028 over the 400 m bands, on a window that cannot be canonically confirmed. The resolution measures the same quantity.

The one thing that would change the picture

Neither study has been able to move the production model, and both have converged on the same explanation: the market price plus 28 columns already contains most of what a new construction from the same archive can supply. The binding constraint is not the estimator — three quite different estimators have now been tried — it is the information set.

The reference project's own backlog names the step that would change this, and neither study has performed it: a **prospective shadow test against timestamped pre-race price snapshots**. That is the only measurement in this problem that is not already contaminated by the market having seen everything the archive contains, and `hkjc.collection.live.odds` already captures the data for it.